Sankalp Jha

Systems Engineer - Kernel, PCIe/DMA Data Paths & Hardware Bring-Up
Delhi NCR, India

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Education

Year	Degree/Certificate	Institute	CPI/%
Oct 2022 – Aug 2026	B.Tech in CS & IT	Ajay Kumar Garg Engineering College, Delhi NCR	**7.25/10**
2021 – 2022	Senior Secondary (XII)	New Era School, Delhi NCR	**91.2%**

Experience

Google Summer of Code 2025 Contributor | *P4 Language Consortium, Stanford, CA* (***May 2025 – Sept 2025***)

SpliDT Project: Scaling Stateful Decision Tree Algorithms in P4 *P4, C++, Python, P4Runtime, Docker, Intel Tofino*

- Architected a switch-native framework compiling ML decision trees directly into P4 data planes using Partitioned Decision Trees (PDTs) with TCAM-efficient Subtree IDs, working around hardware ASIC constraints on Intel Tofino silicon.
- Built a C++/P4Runtime data-path testing pipeline spanning Tofino hardware and Mininet software targets, improving deployment reliability by **30%**.
- Optimized low-level SID recirculation logic to hit **10** Gbps line-rate throughput with up to **5x** lower inference latency than control-plane inference.

Open Source Contributions

- **Project-HAMi/HAMi:** **#2292** fix(device): use len(devices) for GenReason in kunlun and awsneuron; **#2307** fix(device-plugin): release node lock when bind-phase patch fails; **#2259** fix(scheduler): handle ResourceQuota tombstones.
- **WasmEdge/WasmEdge:** **#4470** fix(wasi): resolve **100%** CPU usage on macOS due to clock mismatch; **#4466** feat(docker): add standalone alpine-base image build pipeline.
- **facebook/bpfilter:** **#507** build: update clang-tidy brace config, fix opts.c/set.c, and NOLINT dump.c.

Key Projects

XNIC v1 | *C, Linux Kernel, PCI, DMA, NAPI, QEMU, DPDK (rte_ethdev)* (*Project Link*)

Objective: Developed a clean-room Linux Ethernet driver with robust fault recovery and a user-space DPDK forwarding path.
Approach: Implemented a C driver featuring PCI MMIO, DMA rings, interrupt-driven NAPI, QEMU automation, and fault injection.
Impact: Maintained zero packet loss across **646,400** packets, proving stability through **1,000** interface cycles and reset-under-traffic tests.

Los-Tecnicos | *Go, Rust, Soroban, Raspberry Pi, ESP32, TypeScript* (*Project Link*)

Objective: Meter and tokenize renewable-energy production from a physical sensor mesh rather than from self-reported data.
Approach: Built a Raspberry Pi/ESP32 sensor mesh and the Go backend that ingests it; wrote **4** Soroban smart contracts and drove software/hardware bring-up end to end.
Impact: Won Asia-Pacific Web3 bounties and was recognized by the Government of India.

Do-It | *Go, SSE, HTTP, Raspberry Pi, OpenWrt, Local-first* (*Project Link*)

Objective: Run a private device-to-device sync service directly on constrained home hardware with no cloud dependency.
Approach: Packaged a single static Go SSE/HTTP binary small enough to run on a Raspberry Pi, consumer routers, and repurposed Android handsets.
Impact: Delivers account-free synchronization entirely on local hardware, with no household data leaving the LAN.

Myprod | *Go, Nomad, Traefik, WireGuard, Docker, bare-metal* (*Project Link*)

Objective: Turn a fleet of heterogeneous bare-metal and VPS machines into one schedulable compute pool.
Approach: Built a Go control plane over Nomad scheduling and WireGuard mesh networking, with guarded node lifecycle operations and live per-node CPU/memory telemetry.
Impact: Runs mixed-architecture nodes (x86_64 and ARM64) under a single scheduler with idempotent DNS and ingress management.

Technical Skills

- **Language & RTL:** Verilog, C, C++, Python, P4, Go
- **High-Speed Systems & Interfaces:** PCIe, Ethernet, DPDK, DMA, BAR0 MMIO, NAPI, eBPF
- **Simulation & Verification:** QEMU, Wireshark, Scapy, LLVM, MLIR, Fault Injection, CI/CD
- **Tools & Platforms:** Linux Kernel (x86_64, ARM64), VS Code, Docker, Git

Publications

- R. Bharadwaj, **S. Jha**, V. Malyan, R. Gupta, "Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions," ***2026 ICCSC***, Ghaziabad, India, Feb. **2026**. DOI: 10.1109/ICCSC67078.2026.11468597

Certifications & Honors

- **2x** AKTU State TT Gold (**2023**, **2024**); Robotics Olympiad Bronze, Thailand (**2019**); NASA, NPTEL & ISRO certifications